

Is Semantic Level of Detail Linearly Decodable from Frozen Scientific Text Embeddings?

Introduction

Semantic Level of Detail (SLoD) describes how abstract or detailed a text span is. In other words, it represents the level of abstraction of a particular text excerpt. This experiment differentiates three levels of detail. **Macro** includes broad general contributions, summaries, and conclusions. **Meso** represents section-level framing or overview. **Micro** includes methodological, implementation, experimental, or numerical details.

SLoD matters for the Temporal Knowledge Hypergraph (TKH) because it can help organize extracted knowledge by granularity and provide a retrieval-routing signal toward the appropriate level of detail. The main research question is whether SLoD is linearly decodable from frozen scientific-text embeddings. More specifically, can a simple logistic-regression classifier distinguish macro, meso, and micro spans from frozen SciBERT embeddings under controlled conditions?

Methodology

Weak supervision and datasets

Labels were generated automatically from scientific-document structure instead of manual annotation. Abstracts, the first two introduction paragraphs, and conclusion paragraphs were labelled **macro**. The first sentence of the lead paragraph of eligible sections was labelled **meso**. Non-lead paragraphs in methods, approach, implementation, experiments, results, and evaluation sections were labelled **micro**.

Each span records paper ID, domain, section name, section family, paragraph index, label, label source rule, text, token count, and row ID.

The primary dataset was QASPER, with NLP as the domain. It contained 5,292 spans from 877 papers and exactly 1,764 spans per class. Spans were filtered to 30-300 whitespace tokens. Paper-ID splitting produced 613 training, 87 validation, and 177 test papers, with no overlap.

The external biomedical test used the NCBI PMC Open Access BioC API. It contained 1,500 spans from 197 papers, exactly 500 spans per class, and 30-300 tokens per span. Biomedical data were used only for testing.

Embeddings, probe, and conditions

The model was *allenai/scibert_scivocab_uncased*, selected because it was pretrained on scientific language. The encoder remained completely frozen, and representations used attention-mask-aware mean pooling. The linear probe was logistic regression with balanced class weights. The majority baseline always predicted the most frequent training class and used neither embeddings nor text. The section-name baseline used only `section_name`.

The three conditions were: (1) in-domain QASPER NLP; (2) a separate length-controlled evaluation using the same paper-level split; and (3) cross-domain biomedical evaluation, training on QASPER NLP and testing without adaptation on PMC papers.

Results

Overall performance

Condition	Model	Accuracy	Macro-F1
In-domain	SciBERT probe	0.8858	0.8858
In-domain	Section-name	0.7884	0.7850
In-domain	Majority	0.3155	0.1599
Length-controlled	SciBERT probe	0.7790	0.7803
Length-controlled	Section-name	0.7884	0.7850
Length-controlled	Majority	0.3155	0.1599
Cross-domain biomedical	SciBERT probe	0.8240	0.8237
Cross-domain biomedical	Section-name	0.8300	0.8267
Cross-domain biomedical	Majority	0.3333	0.1667

Condition	Class	Precision	Recall	F1
In-domain	Macro	0.883	0.898	0.890
	Meso	0.892	0.932	0.911
	Micro	0.882	0.831	0.856
Length-controlled	Macro	0.772	0.768	0.770
	Meso	0.833	0.831	0.832
	Micro	0.736	0.742	0.739
Cross-domain biomedical	Macro	0.821	0.814	0.817
	Meso	0.882	0.896	0.889
	Micro	0.768	0.762	0.765

In-domain	Pred macro	Pred meso	Pred micro
True macro	333	15	23
True meso	6	314	17
True micro	38	23	299
Length-controlled	Pred macro	Pred meso	Pred micro
True macro	285	23	63
True meso	24	280	33
True micro	60	33	267
Cross-domain	Pred macro	Pred meso	Pred micro
True macro	407	17	76
True meso	13	448	39
True micro	76	43	381

The probe substantially outperformed both baselines in the normal in-domain condition, suggesting predictive information beyond section names. Macro-F1 decreased from 0.8858 to 0.7803 after length control, demonstrating that span length was a meaningful confound. The controlled probe remained above majority but approximately tied with the section-name baseline. The biomedical probe remained far above the majority but was also approximately tied with section name. Weak SLoD labels were

linearly decodable, while length and section structure remained important confounds.

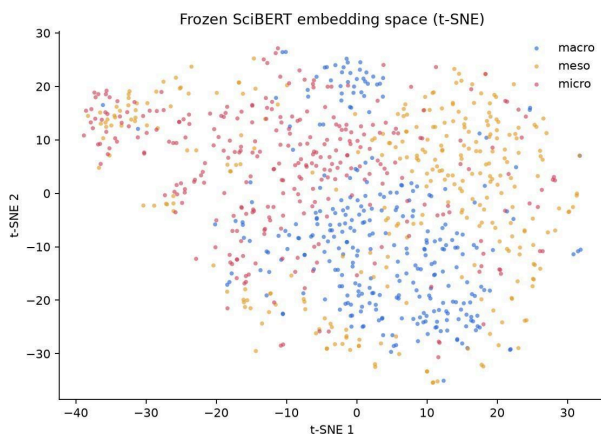
Error Analysis

Micro was the most difficult class to determine. Its F1 was 0.856 in-domain, 0.739 under length control, and 0.765 in the biomedical condition. Micro spans can contain methodology, implementation details, numerical results, interpretations, and broad summaries, making them less linguistically consistent.

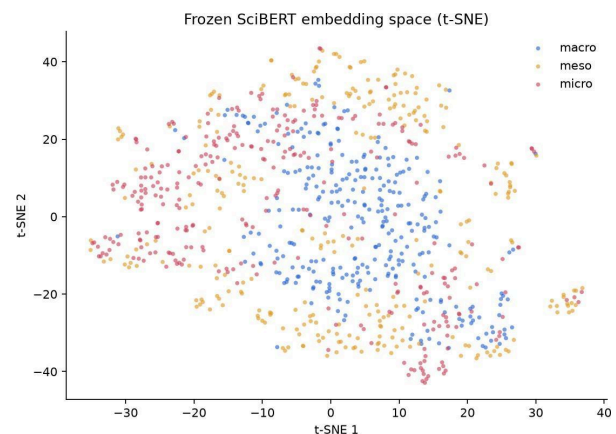
Meso was not the hardest class. It may be recognizable through expressions such as 'in this section, we...', 'we first describe...', and 'the following experiment...'. Its performance may therefore reflect rhetorical cues as well as abstraction.

The largest confusion was between macro and micro. In-domain, 38 micro examples were predicted as macro. Under length control, macro was predicted as micro 63 times and micro as macro 60 times. In the biomedical condition, both directions contained 76 errors. Results sections sometimes summarize findings broadly, while conclusions can contain technical details. A technically detailed conclusion may receive a macro weak label because of its location but be predicted as micro. Some errors may therefore reflect weak-label noise.

Embedding-space visualizations



QASPER test embeddings



PMC biomedical embeddings

The t-SNE visualizations show partial clustering by weak SLoD label.

Discussion and Future Work

SLoD is linearly decodable from frozen scientific-text embeddings under the tested conditions. However, the experiment could not fully eliminate confounds such as section name and document structure. Embeddings predicted the weak labels, but section names were also strongly correlated with them. This relationship may be connected to the language and structure of scientific papers.

The conclusion should therefore remain careful: SLoD is linearly decodable from frozen scientific-text embeddings under these experimental conditions, but the signal may partly depend on scientific-paper structure. The experiment demonstrates decodability, not that the model understands SLoD.

Use as a RAG routing signal

SLoD could help a RAG system decide what level of detail to retrieve. If a query requires macro-level information, the system could retrieve macro spans and generate a high-level answer. If

the user needs technical or numerical details, it could retrieve micro spans instead. This would help return the type of information that the user actually needs.

Future work

Future work should first obtain human-annotated SLoD labels to distinguish true abstraction level from structural patterns more precisely. SciBERT could then be compared with general BERT and other embedding models. The probe should also be tested on non-scientific documents to measure dependence on scientific-writing conventions. Continuous SLoD regression could be investigated instead of three discrete classes.

Finally, further experiments should determine whether SLoD is generally linearly decodable from frozen embeddings or whether the observed signal depends specifically on scientific-paper structure.